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MA3521 — In-Class Worksheet — Monday
Do not row reduce or solve.

Name:

1. $\mathbf{y}' = \boldsymbol{A}\mathbf{y} + e^{-2t}\begin{pmatrix}-3\\-5\\-1\end{pmatrix}$ $\quad \boldsymbol{A} = \begin{pmatrix}1 & -2 & 0\\ 2 & 1 & 0\\ 0 & 0 & -1\end{pmatrix}$ $\quad \mathbf{y}(0) = \begin{pmatrix}2\\3\\1\end{pmatrix}$

   Eigenpairs $1+2i,\ \begin{pmatrix}1\\-i\\0\end{pmatrix}$ $\quad 1-2i,\ \begin{pmatrix}1\\i\\0\end{pmatrix}$ $\quad -1,\ \begin{pmatrix}0\\0\\1\end{pmatrix}$

   (a) $\mathbf{y}_c(t) =$

   (b) A suitable UC form is

   $\mathbf{y}_p(t) =$

   (c) To determine the UC coefficients you would row reduce

   $\boldsymbol{M}_{\text{UC}} =$

   (d) To satisfy the IC you would row reduce

   $\boldsymbol{M}_{\text{IC}} =$

Not on exam $\mathbf{y}(t) =$

2. $\mathbf{y}' = \boldsymbol{A}\mathbf{y} + \cos(2t)\begin{pmatrix}-3\\-5\\-1\end{pmatrix}$ $\quad \boldsymbol{A} = \begin{pmatrix}-1 & -2 & 1\\-2 & -3 & 3\\-1 & -3 & 2\end{pmatrix}$ $\quad \mathbf{y}(0) = \begin{pmatrix}2\\3\\1\end{pmatrix}$

Eigenpairs i, $\begin{pmatrix}1\\-i\\1-i\end{pmatrix}$ $\quad -i$, $\begin{pmatrix}1\\i\\1+i\end{pmatrix}$ $\quad -2$, $\begin{pmatrix}1\\1\\1\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$

(b) A suitable UC form is

$\mathbf{y}_p(t) =$

(c) To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$

(d) To satisfy the IC you would row reduce

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